

Master's Comprehensive Examination

Part II

Applied (90 minutes)

March 10, 2001

Instructions: Answer three out of four questions. Open book and open notes.

1. A study was conducted to investigate some effects of physical training. Sample data (weights in kilograms) are listed below. (See "Effect of Endurance Training on Possible Determinants of VO_2 During Heavy Exercise," by Casaburi et al., *Journal of Applied Physiology*, Vol. 62, No. 1.)

- (a) Use the sign test to test the claim that the training has no effect on weight.
- (b) Do this problem using the Wilcoxon signed-rank test.

Pre-training	99	57	62	69	74	77	59	92	70	85
Post-training	94	57	62	69	66	76	58	88	70	84

2. The table below classifies a sample of psychiatric patients by their diagnosis and by whether their treatment prescribed drugs.

Diagnosis	Drugs	No Drugs
Schizophrenia	105	8
Affective disorder	12	2
Neurosis	18	19
Personality disorder	47	52
Special symptoms	0	13

- (a) Report the P-value for a test of independence and interpret the result.
- (b) Partition chi-squared into three components to describe differences and similarities among the diagnoses, by comparing
 - (i) the first two rows,
 - (ii) the third and fourth rows,
 - (iii) the last row, the first and second rows combined, and the third and fourth rows combined.

3. Consider an experiment with four factors, A, B, C, and D, each at two levels.

- (a) Write out the treatment combinations in this experiment.
- (b) Assuming there is only one observation per treatment, outline an ANOVA for this example and describe and justify what you would use for an error term to test for significance of various factors and interactions.
- (c) Explain how you would determine the AD interaction from the responses to the treatment combination given in part (a).

4. Shown below are the number of galleys for a manuscript (X) and the total dollar cost of correcting typographical errors (Y) in a random sample of recent orders handled by a firm specializing in technical manuscripts. Since Y involves variable costs only, an analyst wished to determine whether regression-through-the-origin model (1) is appropriate for studying the relation between the two variables.

i :	1	2	3	4	5	6	7	8	9	10	11	12
X_i :	7	12	10	10	14	25	30	25	18	10	4	6
Y_i :	128	213	191	178	250	446	540	457	324	177	75	107

Note: Model (1) is a straight line through the origin, with slope β_1 .

$$Y_i = \beta_1 X_i + \epsilon_i$$

where:

β_1 is a parameter,

X_i are known constants,

ϵ_i are independent $N(0, \sigma^2)$.

(a) Fit regression model (1) and state the estimated regression function.

(b) Plot the estimated regression function and the data. Does a linear regression function through the origin appear to provide a good fit here? Comment.

(c) In estimating costs of handling prospective orders, management has used a standard of \$17.50 per galley for the cost of correcting typographical errors. Test whether or not this standard should be revised; use $\alpha = .02$. State the alternatives, decision rule, and conclusion.

(d) Obtain a prediction interval for the correction cost on a forthcoming job involving 10 galleys. Use a confidence coefficient of 98 percent.